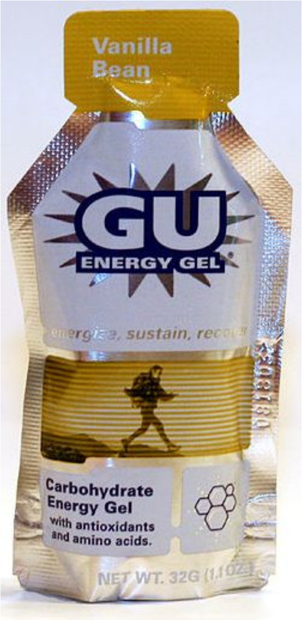
Energy gels and race times

MATH 213

Group 1 - Do energy gels make you run faster?



Consider:

“My friend tried using an energy gel before a 10K and then ran her fastest time ever.”

1. Does this provide convincing evidence that energy gels cause people to run faster? Why or why not?

   No, because it’s anecdotal and incidental. There’s no way to properly say whether or not it was the gels or if it was something the friend was doing outside of the gels.

Now instead, suppose you conduct a study where you meet 1000 runners at the startline for a race. You interview each runner and record whether each person has used an energy gel or not and then, at the end of the race, record their finishing time.

2. What type of study is this? (Observational / Experimental?)

Observational

3. What are the variables? Label as explanatory and response.

Consumed energy gel (explanatory), runner time (response)

4. If it turns out that the people who used energy gels have faster race times, does this provide convincing evidence that the energy gels caused them to have faster race times?

I’d say, while it’s more convincing, it still probably doesn’t because there are way too many free variables to take into consideration that are not accounted for.

5. If you answered YES to the previous question, explain why. If NO, provide a possible confounding variable that could explain the association between energy gels and faster race times. Be sure to explain how the confounding variable is associated with both increased energy gel use and with faster race times.

One confounding variable, how much they trained, because people who train more will of course be faster, and it’s likely the ones who train more simply figured out that the drink was a better fuel source because those who are more athletic actively do burn more calories, and thus need a higher intake of food and sugar.

Group 2 - Do energy gels make you run faster?

Consider:

"My friend tried using an energy gel before a 10K and then ran her fastest time ever."

1. Does this provide convincing evidence that energy gels cause people to run faster? Why or why not?

    This does not provide convincing evidence because: It does not include other variables such as rest, diet, or a comparative time. The evidence is also anecdotal, and only has one case in the sample.

Now instead, suppose you conduct a study where you meet 1000 runners at the startline for a race. You interview each runner and record whether each person has used an energy gel or not and then, at the end of the race, record their finishing time.

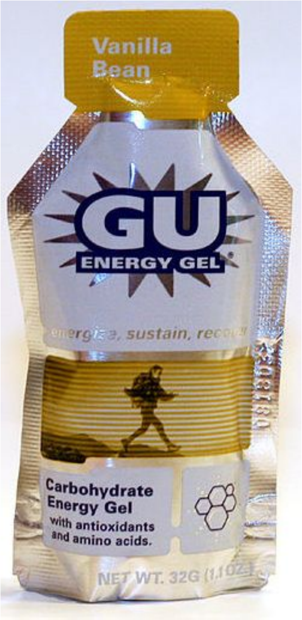
2. What type of study is this? (Observational / Experimental?) **Observational**

3. What are the variables? Label as explanatory and response. **Energy gel usage (explanatory), finishing time (response)**

4. If it turns out that the people who used energy gels have faster race times, does this provide convincing evidence that the energy gels caused them to have faster race times? **No. There could be a correlation but its not necessarily causal.**

5. If you answered YES to the previous question, explain why. If NO, provide a possible confounding variable that could explain the association between energy gels and faster race times. Be sure to explain how the confounding variable is associated with both increased energy gel use and with faster race times. **Someone whos more athletic may be more likely to drink energy gel, meaning they would have finished faster either way.**

Group 3 - Do energy gels make you run faster?



Consider:

“My friend tried using an energy gel before a 10K and then ran her fastest time ever.”

1. Does this provide convincing evidence that energy gels cause people to run faster? Why or why not?

   No, there are a number of outside factors that could contribute to a faster overall time, such as adrenaline, amount of sleep, etc..

Now instead, suppose you conduct a study where you meet 1000 runners at the startline for a race. You interview each runner and record whether each person has used an energy gel or not and then, at the end of the race, record their finishing time.

2. What type of study is this? (Observational / Experimental?)

Observational

3. What are the variables? Label as explanatory and response.

1) male/female
2) Consumption of energy gel (explanatory)
3) Finishing time (response)

4. If it turns out that the people who used energy gels have faster race times, does this provide convincing evidence that the energy gels caused them to have faster race times?

No observational studies can discover association but can not prove causality

5. If you answered YES to the previous question, explain why. If NO, provide a possible confounding variable that could explain the association between energy gels and faster race times. Be sure to explain how the confounding variable is associated with both increased energy gel use and with faster race times.

Confounding variable:

1) adrenaline, the collective group consuming gels may cause the individual to consume more energy gels, the adrenaline may cause a runner to run the first few minutes faster leading to a faster time.

2) Race environment

Group 4 - Do energy gels make you run faster?

Consider:

“My friend tried using an energy gel before a 10K and then ran her fastest time ever.”

1. **Does this provide convincing evidence that energy gels cause people to run faster? Why or why not?**

   No, because there is only one case which isn't a good representation of the population

Now instead, suppose you conduct a study where you meet 1000 runners at the startline for a race. You interview each runner and record whether each person has used an energy gel or not and then, at the end of the race, record their finishing time.

2. What type of study is this? (Observational / Experimental?)

Observational study

3. What are the variables? Label as explanatory and response.

Explanatory: Used energy gel or not

Response: Finishing time

4. If it turns out that the people who used energy gels have faster race times, does this provide convincing evidence that the energy gels caused them to have faster race times?

Not necessarily

5. If you answered YES to the previous question, explain why. If NO, provide a possible confounding variable that could explain the association between energy gels and faster race times. Be sure to explain how the confounding variable is associated with both increased energy gel use and with faster race times.

A possible confounding variable is athleticism; those who have higher athletic ability might be more likely to use energy gel and have faster race times

Group 5 - Do energy gels make you run faster?

Consider:

“My friend tried using an energy gel before a 10K and then ran her fastest time ever.”

1. **Does this provide convincing evidence that energy gels cause people to run faster? Why or why not?**

No. There needs to be more cases in a study for it to provide a convincing argument.

Now instead, suppose you conduct a study where you meet 1000 runners at the startline for a race. You interview each runner and record whether each person has used an energy gel or not and then, at the end of the race, record their finishing time.

2. What type of study is this? (Observational / Experimental?)

Observational

3. What are the variables? Label as explanatory and response.

Energy gel usage - explanatory

Finishing time - response

4. If it turns out that the people who used energy gels have faster race times, does this provide convincing evidence that the energy gels caused them to have faster race times?

No

5. If you answered YES to the previous question, explain why. If NO, provide a possible confounding variable that could explain the association between energy gels and faster race times. Be sure to explain how the confounding variable is associated with both increased energy gel use and with faster race times.

No because the study doesn't consider their previous race times meaning we don't know whether the energy gel makes an improvement to their finishing time. A possible confounding variable that could explain the association is their personal best time. It is possible that the people who had used energy gels were just normally faster athletes.

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Group 6 - Do energy gels make you run faster?

Consider:

“My friend tried using an energy gel before a 10K and then ran her fastest time ever.”

1. Does this provide convincing evidence that energy gels cause people to run faster? Why or why not?

Now instead, suppose you conduct a study where you meet 1000 runners at the startline for a race. You interview each runner and record whether each person has used an energy gel or not and then, at the end of the race, record their finishing time.

2. What type of study is this? (Observational / Experimental?)

3. What are the variables? Label as explanatory and response.

4. If it turns out that the people who used energy gels have faster race times, does this provide convincing evidence that the energy gels caused them to have faster race times?

5. If you answered YES to the previous question, explain why. If NO, provide a possible confounding variable that could explain the association between energy gels and faster race times. Be sure to explain how the confounding variable is associated with both increased energy gel use and with faster race times.

Group 7 - Do energy gels make you run faster?

Consider:

“My friend tried using an energy gel before a 10K and then ran her fastest time ever.”

1. Does this provide convincing evidence that energy gels cause people to run faster? Why or why not?

   No because it is too little information and There are better tests to help prove or disprove the effect of energy gel on speed.

Now instead, suppose you conduct a study where you meet 1000 runners at the startline for a race. You interview each runner and record whether each person has used an energy gel or not and then, at the end of the race, record their finishing time.

2. What type of study is this? (Observational / Experimental?)

Observational

3. What are the variables? Label as explanatory and response.

Gel- Explanatory

Finish Time- Response

4. If it turns out that the people who used energy gels have faster race times, does this provide convincing evidence that the energy gels caused them to have faster race times?

Yes, if the data isn’t skewed

5. If you answered YES to the previous question, explain why. If NO, provide a possible confounding variable that could explain the association between energy gels and faster race times. Be sure to explain how the confounding variable is associated with both increased energy gel use and with faster race times.

If the group that took the gel ran faster race times then yes it proves that the gel caused those faster race times factoring in one group does not have confounding variables. These confounding variables could include athleticism, training, rest, adrenaline

Group 8 - Do energy gels make you run faster?

Consider:

“My friend tried using an energy gel before a 10K and then ran her fastest time ever.”

1. Does this provide convincing evidence that energy gels cause people to run faster? Why or why not?

No, because there is a sample size of one and there is no way to determine causation, as there is no control group.

Now instead, suppose you conduct a study where you meet 1000 runners at the startline for a race. You interview each runner and record whether each person has used an energy gel or not and then, at the end of the race, record their finishing time.

2. What type of study is this? (Observational / Experimental?)

3. What are the variables? Label as explanatory and response.

- Runners time (response)
- Did the runner use an energy gel? (explanatory)

4. If it turns out that the people who used energy gels have faster race times, does this provide convincing evidence that the energy gels caused them to have faster race times?

- If evidence is convincing that the energy gel causes faster run times, it strongly suggests a correlation between the two variables, however; since this is not an experiment, but an observational study, it does not necessarily provide convincing evidence of CAUSATION. There would need to be further studies performed to eliminate confounding variables that are present in this observational study.

5. If you answered YES to the previous question, explain why. If NO, provide a possible confounding variable that could explain the association between energy gels and faster race times. Be sure to explain how the confounding variable is associated with both increased energy gel use and with faster race times.

- People who invest money in energy gels may already have a higher aptitude at running than those who don’t, which would seemingly inflate the effectiveness of the gels.

Group 9 - Do energy gels make you run faster?

Consider:

“My friend tried using an energy gel before a 10K and then ran her fastest time ever.”

1. Does this provide convincing evidence that energy gels cause people to run faster? Why or why not?

This does not provide convincing evidence. It could be a result of the placebo effect.

Now instead, suppose you conduct a study where you meet 1000 runners at the startline for a race. You interview each runner and record whether each person has used an energy gel or not and then, at the end of the race, record their finishing time.

2. What type of study is this? (Observational / Experimental?)

Observational study

3. What are the variables? Label as explanatory and response.

Finishing time - response

Usage of energy gel - explanatory

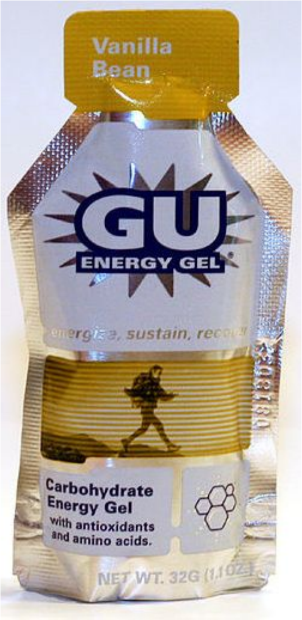
4. If it turns out that the people who used energy gels have faster race times, does this provide convincing evidence that the energy gels caused them to have faster race times?

Evidence is provided but is spectulatory as the runners who used the energy gel could all be experiencing the gel as a placebo. The two variables are associated, but the evidence does not convince of that they are causally associated.

5. If you answered YES to the previous question, explain why. If NO, provide a possible confounding variable that could explain the association between energy gels and faster race times. Be sure to explain how the confounding variable is associated with both increased energy gel use and with faster race times.

Amount of training; athleticism as athletic people will run faster and are more likely to use energy gel; conformist maybe more likely take energy gels with people around them and may try harder to not fall behind pack

Group 10 - Do energy gels make you run faster?



Consider:

"My friend tried using an energy gel before a 10K and then ran her fastest time ever."

1. Does this provide convincing evidence that energy gels cause people to run faster? Why or why not?
    - Anecdotal Evidence is a normal that stands because of its unusualness.

Now instead, suppose you conduct a study where you meet 1000 runners at the startline for a race. You interview each runner and record whether each person has used an energy gel or not and then, at the end of the race, record their finishing time.

2. What type of study is this? (Observational / Experimental?)

Observational

3. What are the variables? Label as explanatory and response.

Categorical, Do they use an energy gel or not?

Numerical, What was their finishing time and where does that place them in the ranking?

4. If it turns out that the people who used energy gels have faster race times, does this provide convincing evidence that the energy gels caused them to have faster race times?

No, they might simply have already been better athletes and know that the energy gel will have nutrients their body will need after the race.

5. If you answered YES to the previous question, explain why. If NO, provide a possible confounding variable that could explain the association between energy gels and faster race times. Be sure to explain how the confounding variable is associated with both increased energy gel use and with faster race times.

How many races have they competed in prior to this race? Ordinal ranking to show possible correlation with energy gel usage and racing experience